

Sajeevan Veeriah



Robotics, Mechatronics, AI/ML & End-To-End Automation Engineer

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PROFILE

Mechatronics engineer (Honours, Distinction) who takes systems from requirements to handover across robotics, industrial automation, embedded hardware and software. Experience spans PLC/SCADA integration in GMP manufacturing, ADAS and vehicle-software validation, emissions and EV testing, and LoRaWAN telemetry. Now delivering client electronics, firmware and applications through SV Tech Solutions.

Core strengths: Robotics and autonomy | Industrial automation and controls | Embedded systems and PCB design | Vehicle test and validation | Software, AI/ML and data | Commissioning, quality and compliance

TECHNICAL SKILLS

Automation and controls: PLC logic and integration, Siemens TIA Portal, HMI/SCADA (iFIX, PVI+), drives, MES, Modbus, PROFINET, industrial IT/OT, FAT/SAT, commissioning, GMP, ITPs, MDRs and AS/NZS compliance.

Robotics and embedded: ROS 2, Nav2, SLAM, EKF sensor fusion, Gazebo, RViz, C/C++, ESP32/ESP32-S3, STM32, FreeRTOS, CAN/CAN FD, UART, I2C, SPI and BLE.

Software, AI and data: Python, MATLAB/Simulink, TypeScript, React, Rust (Tauri), SQL/SQLite, Git, CI/CD, OpenCV, scikit-learn and local language-model retrieval, memory and evaluation workflows.

Hardware, IoT and test: Altium, KiCad, SolidWorks, Fusion 360, PCB and enclosure design, 3D printing, Linux, LoRaWAN, ChirpStack, MQTT, InfluxDB, Grafana, CANoe, CANalyzer and calibrated data acquisition.

PROFESSIONAL EXPERIENCE

Independent Consultant Engineer

Jun 2026 - Present

SV Tech Solutions

- Deliver client systems from requirements through electronics, firmware, software, testing and release.
- **Gendio:** designed a four-layer ESP32-S3 controller, serial-adaptor PCB, display firmware and browser configuration; extended vehicle-position sensing and prepared fabrication files. Assembled-system and field validation pending.
- **Stan Wootton Locksmiths:** sole engineer on pricing and inventory software (React/TypeScript, Tauri/Rust, SQLite), including operator-reviewed ServiceM8 imports with change, exception, rollback and audit reports.

Automation and Controls Engineer

Jan 2026 - Jun 2026

JAG Process Solutions

- Integrated PLC logic, HMI/SCADA, field devices, drives and production data for pharma, biotech and food plants.
- Migrated validated iFIX content to PVI+ and completed functional checks; supported FAT/SAT, commissioning, GMP handover and fault tracing on MiR mobile robots.

Product Development Test Engineer (Contract)

Oct 2025 - Jan 2026

Ford Motor Company via Invenio

- Tested vehicle-software integration and ADAS features through feature-vehicle, breadboard and regression testing.
- Instrumented vehicles and analysed CAN/CAN FD data in CANoe and CANalyzer to isolate faults and document software-readiness evidence.

Technical Officer & Quality Assurance

Jul 2024 - Aug 2025

ABMARC

- Conducted emissions, EV/PHEV range and vehicle-system tests with calibrated instrumentation and DAQ.
- Maintained traceable test, safety and quality records for certification review.

Consultant Engineer, IoT and Projects Administrator

Feb 2024 - Aug 2024

DuxTel

- Built ESP32 and LoRaWAN AU915 telemetry using ChirpStack, MQTT, InfluxDB and Grafana.
- Integrated custom PCBs, CAN/GPS, MikroTik connectivity and Linux services for field trials and handover.



MANUFACTURING AND QUALITY EXPERIENCE

Production Line Worker / Supervisor / Commissioning Support; Cellar Hand

Aug 2022 - Feb 2024

IDL Australia

- Led output, changeovers and handovers; maintained quality and traceability and supported canning-line commissioning to Sep 2023.
- From Sep 2023, supported cellar sanitation, sampling and production records.

Production Line Operator and Commissioning Support

Sep 2021 - Dec 2021

Carbon Revolution

- Operated carbon-fibre wheel equipment and maintained quality records; supported equipment trials, setup and first-level fault recovery.

Undergraduate Quality Engineer

Mar 2018 - Mar 2020

Thornton Engineering Australia

- Supported drawing review, ITPs, MDRs, material traceability and AS/NZS compliance in structural-steel fabrication.

SELECTED ENGINEERING PROJECTS

Full portfolio: sv.sajeevanveeriah.workers.dev/work

Autonomous navigation rover | [ROS 2 autonomy](#)

Built a differential-drive platform with LiDAR and IMU; integrated SLAM, EKF localisation and Nav2 in C++/Python and verified navigation and recovery in Gazebo and RViz. Autonomy validated in simulation.

Ataxia assessment device | [Embedded mechatronics](#)

Designed an ESP32 prototype, Altium PCB and enclosure with four Hall-effect channels at 100 Hz and Bluetooth reporting; tested accuracy, reversals, drift and temperature behaviour in MATLAB.

SPC-001 waterless solar-panel cleaner | [Robotics](#)

Developed mechanical CAD, electronics, supervisory control and coverage planning. Wrote an independent research paper on contact mechanics, traction and energy that distinguishes analysis from physical performance.

Deadline-aware robot runtime assurance | [Simulation research](#)

Modelled sensor uncertainty, measurement age and command delay in a braking supervisor; compared recovery strategies using fault-injection benchmarks.

VeerAI | [Local AI system](#)

Built a locally deployed language-model system with document ingestion, retrieval, memory, tools and evaluation.

Potato-processing automation | [Industrial simulation](#)

Built a browser-based process simulation informed by Farm Frites discussions, with control and operator-interface concepts. Uses simulated data; no factory implementation is claimed.

Community platforms and engineering tools | [Full-stack web](#)

Built the NDCC website and Dino Coach fantasy-cricket platform, including CMS, registration, teams, budgets, payments and meal ordering. Developed Engineering Mastery Lab for technical learning and calculations.

EDUCATION

Bachelor of Mechatronics Engineering (Honours), Distinction

2025

Deakin University

Higher National Diploma in Mechatronics, Robotics and Automation Engineering, Distinction

2016

Cardiff Metropolitan University

MEMBERSHIP AND LANGUAGES

Member, Engineers Australia | English, Tamil and Sinhala | Community leadership and engineering mentoring.